M25 Methoxyflurane (Penthrox)

Scope of Practice	EMT
Classification	• Inhaled Anesthetic
Usual Presentation	• 3ml vial
Mechanism of Action	• Depresses the central nervous system primarily through enhancing GABA-A receptor activity to increase inhibitory neurotransmission. • Inhibits NMDA receptors to reduce excitatory pain signals.
Indications	• Moderate to Severe Pain • Moderate Pain with occasional Severe Pain associated with procedures
Contraindications	• Patients unable to obey commands, or are unable to self-administer • History of malignant hyperthermia (rare) • Known renal impairment • Severe liver disease • Hypersensivity to anesthetics
Precautions	• Age >75years – elderly patients are more likely to have renal impairment. • Pre-eclampsia – renal impairment is likely in pre-eclampsia.
Side Effects	• Dizziness • Headache • Mild and temporary sedation
Adult Dose & Administration	• 3mls Methoxyflurane self administered (repeat dose once if required) • Adult maximum dose in 24hours is 6ml (2 doses)
Pediatric Dose & Administration	• Pediatric Maximum dose 3ml (single dose) • Generally, pediatrics should be greater than 5 years old to understand how to self-administer methoxyflurane. • Administration in pediatrics has not been widely studied. However, methoxyflurane is widely administered in the pre-hospital setting with minimal reported adverse effects. • When administered to pediatrics, methoxyflurane should be administered sparingly and more appropriate analgesics should be used once they are available.
Use in Pregnancy and Lactation	• Pregnancy: Safety has not been established. Administration during labor is not recommended. • Lactation: Safety has not been established. If essential to be administered, the mother should be advised to stop breast feeding until advised by their physician.
Additional Information	• Methoxyflurane is an effective analgesic provided adequate coaching is provided. • The analgesic effect of methoxyflurane will wear off approximately 30 seconds after inhalation has stopped. Constant inhalation is required to provide on-going analgesia.